

CSCD 618/DSCD 604: Deep Learning and Neural Networks

Sample Quiz Questions

Each question has **two** correct answers. Pick the **two** correct options. **You are to get both options correct to earn the mark assigned to the question. Having one answer correct and the other wrong earns 0 marks.**

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Question 1 of 10

1 Points

Which shape pairs can be multiplied with NumPy's elementwise $*$ operation?

- ☐ A. v has shape $(1, 5)$ and u has shape $(1, 50)$.
- ☐ B. v has shape $(5, 1)$ and u has shape $(5, 1)$.
- ☐ C. v has shape $(2, 3)$ and u has shape $(3, 2)$.
- ☐ D. v has shape $(50, 5)$ and u has shape $(50, 5)$.

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Question 2 of 10

1 Points

For $z = \tanh(x)$, which statements correctly describe gradient propagation through the activation?

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

- ☐ A. The local derivative is $1 - z^2$ and lies in $[0, 1]$.
- ☐ B. The backpropagated gradient magnitude cannot exceed the upstream gradient magnitude.
- ☐ C. The local derivative is always greater than 1.
- ☐ D. Tanh always amplifies gradients in saturated regions.

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Question 3 of 10

1 Points

Assume the vectors $\mathbf{v}$ and $\mathbf{u}$ are defined as NumPy arrays. The vectors are multiplied together using the $*$ operator: $\mathbf{v} * \mathbf{u}$. Which of the following shapes are valid, meaning the operation will succeed without errors?

- ☐ A. $\mathbf{v} = (1, 5)$ and $\mathbf{u} = (1, 50)$.
- ☐ B. $\mathbf{v} = (1, 5)$ and $\mathbf{u} = (50, 5)$.
- ☐ C. $\mathbf{v} = (1, 5)$ and $\mathbf{u} = (5, 1)$.
- ☐ D. $\mathbf{v} = (1, 1)$ and $\mathbf{u} = (50, 5)$.

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Question 4 of 10

1 Points

Why is i.i.d. sampling desirable when estimating a mini-batch gradient?

- ☐ A. It supports an empirical gradient estimate with lower bias on average.
- ☐ B. It eliminates the need to shuffle training data.
- ☐ C.
Strong correlation among batch examples can bias the gradient estimate and impair learning.
- ☐ D.
It guarantees that every mini-batch gradient equals the full-batch gradient.

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Question 5 of 10

1 Points

Which statements about the vanishing-gradient problem are correct?

☐ A.

Xavier initialization can help reduce vanishing-gradient problems in shallow networks.

☐ B.

Vanishing gradients necessarily make deeper layers learn faster than earlier layers.

☐ C.

Leaky ReLU is less likely than sigmoid to suffer from vanishing gradients.

☐ D.

Tanh never suffers from vanishing gradients.

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Question 6 of 10

1 Points

Two similarly sized binary-classification datasets are combined, one mostly positive and the other entirely negative. Which training choices could make the model fail to learn both classes?

☐ A.

The model may converge toward predicting the negative class for nearly every input.

☐ B.

Training first on positive-heavy batches and then only on negative batches can cause forgetting.

☐ C.

Randomly mixing examples from both datasets in each mini-batch necessarily causes failure.

☐ D.

Using both datasets always produces a perfectly balanced training signal.

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Question 7 of 10

1 Points

An input volume of shape **(10, 10, 3)** is connected either to a fully connected layer with 2 neurons and biases or to a convolutional layer with three **2 × 2** filters and biases. Which parameter counts are correct?

- ☐ A.
The convolutional layer has more trainable parameters than the fully connected layer.
- ☐ B. The convolutional layer has 39 trainable parameters (weights and biases).
- ☐ C.
The fully connected layer has 602 trainable parameters (weights and biases).
- ☐ D. The two layers differ by 566 trainable parameters (weights and biases).

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Question 8 of 10

1 Points

A trained logistic-regression model has no bias and uses weight vector **W** . Which statements remain true after replacing **W** with **$W/2$** ?

- ☐ A.
The test-set class predictions and accuracy are unchanged under the usual **0.5** threshold.
- ☐ B. The sign of **$W^T x$** is unchanged for every example.
- ☐ C. The decision boundary rotates by **90°**.
- ☐ D. Every predicted probability remains numerically identical.

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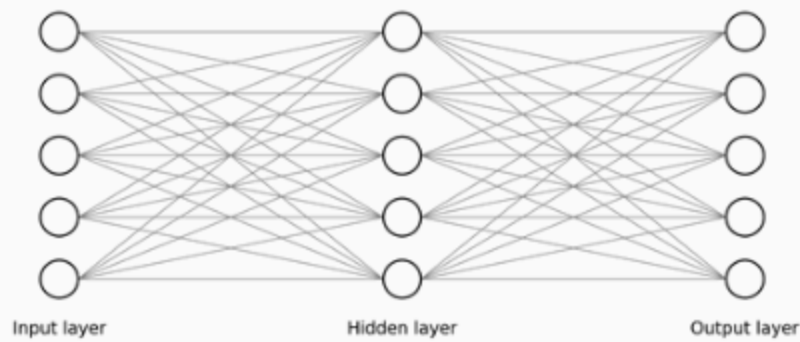


Figure 1. Fully connected network with 5 input units, 5 hidden units, and 5 output units.

For the fully connected network in Figure 1, which statements correctly describe its trainable parameters when biases are included for the hidden and output layers?

- ☐ A. The network contains 50 trainable weights, if all biases are set to false.
- ☐ B. The network contains only 5-bias parameters.
- ☐ C. The network has 60 trainable parameters (weights and biases) in total.
- ☐ D. The network has 75 trainable parameters (weights and biases) in total.

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For the convolutional layer with three 2×2 filters, stride 2, and zero padding applied to an input of shape $(10, 10, 3)$, which statements correctly describe the output feature map?

- ☐ A. The output contains 2 channels.
- ☐ B. The output shape is $(5, 5, 3)$.
- ☐ C. The output shape is $(4, 4, 3)$.
- ☐ D. The output feature map contains 75 real values.

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